

EduTrack School ERP

Complete Multi-Tenant School ERP & Operations Management Suite

Version: 1.0.0 Stable Release

Database Engine: AWS RDS PostgreSQL v15

Configured Branding: Oakridge International School

Prepared By: Covenant Synergy Private Limited

Date: July 2026



PART I

Project Overview & Application Setup Workflow

Project Overview & Core Objective

SaaS Capabilities

Concept: Multi-tenant cloud ERP scaling across hundreds of schools with full data isolation.

Core Goal: Eliminate admin overhead, secure financials, and deliver fast performance insights.

Frontend: Next.js 14, Tailwind CSS, Axios, Framer Motion.

Backend: NestJS, Prisma ORM, JWT auth, AWS S3, PostgreSQL.

Infra: AWS RDS, EC2, Route 53 wildcard resolver, Vercel CDN.

Infrastructure Architecture

Subdomain Routing: Route 53 maps [school].domain to tenant context via Next.js headers.

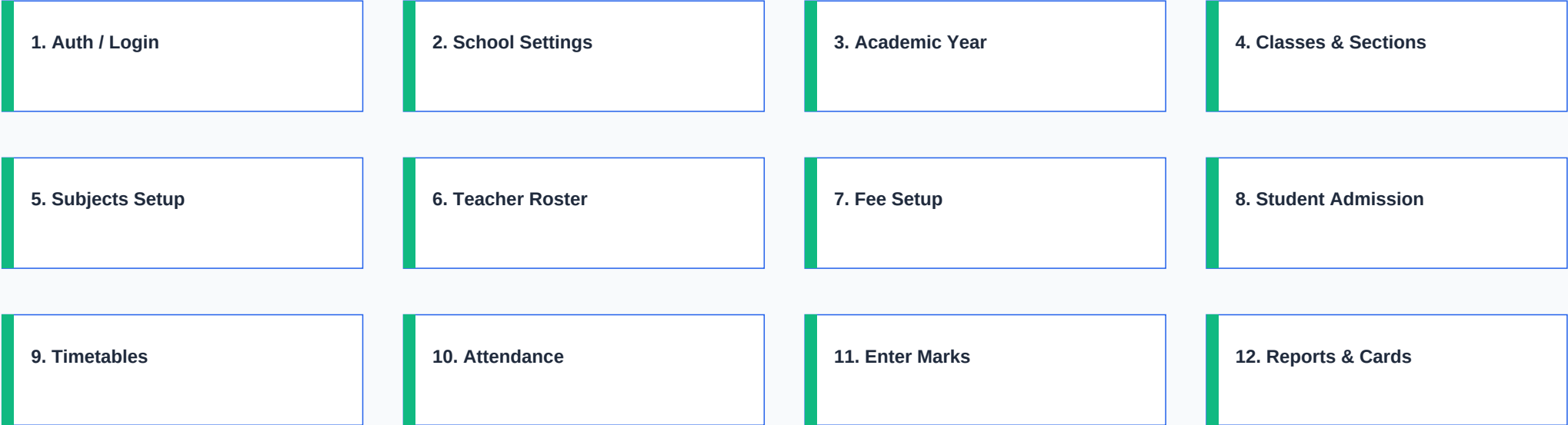
Tenant Isolation: NestJS enforces tenantId on every DB query -- zero data leakage.

Connection Pools: Prisma pools prevent AWS RDS exhaustion under load.

JWT Security: 9999-day JWT in dev; production uses rotating signed tokens.

Core Application Workflow & Setup Order

EduTrack requires: Academic Year -> Classes -> Subjects -> Teachers -> Fee Books -> Student Admission. All FK dependencies must be satisfied before student onboarding.





PART II

Functional Modules Walkthrough & Live Screenshots

Interactive Operations Dashboard

Purpose

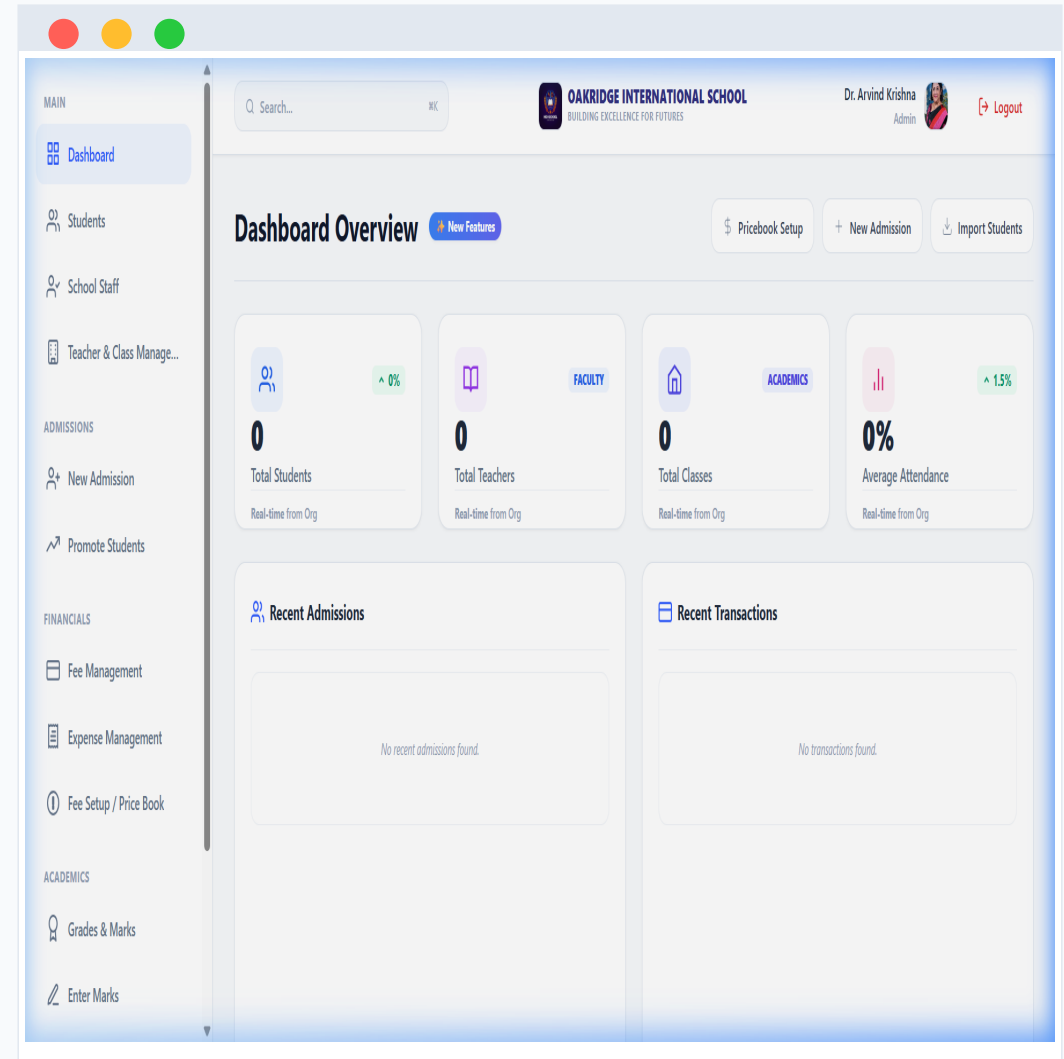
High-level operational overview of school metrics, financials, and headcount.

Key Features

Real-time KPIs: 50 students, 7 teachers, cashflow charts, quick-actions.

Benefits

Instantly spot pending tasks, outstanding invoices, and attendance rates at a glance.



Dashboard: Real screenshot captured from live application.

Onboarding & Checklist System

Purpose

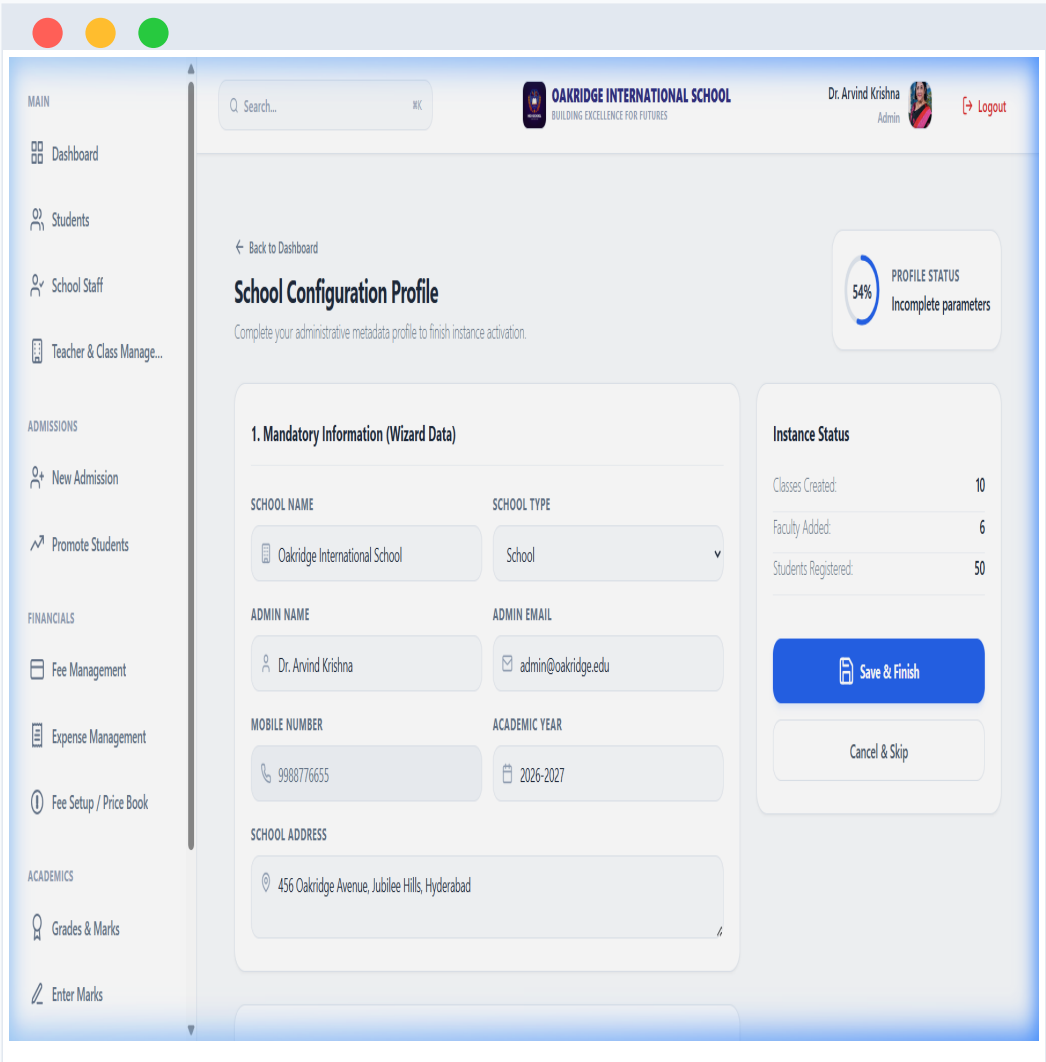
Step-by-step wizard guiding admins through core configuration before student enrollment.

Key Features

Visual checklist for Academic Year, Section Mapping, Schedules, Subjects, and Billing setup.

Benefits

Ensures no FK dependencies are missed during onboarding ? prevents system errors.



School Setup: Real screenshot captured from live application.

System Parameters Configuration

Purpose

Manage global metadata: school contact info, default term, and active academic year.

Key Features

Tenancy details, address config, interactive state controls, and theme properties.

Benefits

Branding metadata flows to receipt layouts, login screens, and report cards dynamically.

The screenshot displays the 'School Setup & Settings' interface. On the left is a sidebar menu with categories: MAIN (Dashboard, Students, School Staff, Teacher & Class Manage...), ADMISSIONS (New Admission, Promote Students), FINANCIALS (Fee Management, Expense Management, Fee Setup / Price Book), and ACADEMICS (Grades & Marks, Enter Marks). The main content area is titled 'School Setup & Settings' and includes a description: 'Replicates 'schoolSetup' LWC. Configure school profiles metadata, active terms, bank account indices, and UPI payments gateways.' Below this are tabs for 'School Profile', 'Bank Registry', 'UPI Gateways', and 'Academic Terms'. The 'School Profile' tab is active, showing the 'Organization Branding & Details' form. The form contains fields for Organization Name (Oakridge International School), School Subtitle / Motto, Tenant Subdomain (oakridge-international-scl.edutrack.com), Contact Email (admin@oakridge.edu), Help Desk Phone (9988776655), and School Logo (Choose File). A 'Branding Preview' section on the right shows the school's profile with the logo and contact details. At the bottom right, there is a 'Developer System Settings' link.

Settings: Real screenshot captured from live application.

Academic Architecture Management

Purpose

Configure classes (Grade 1?5) and map their sections and subjects.

Key Features

Class creator forms, section mappings, subject lists, and teacher workload indicators.

Benefits

Builds the foundational hierarchy driving attendance sheets and exam mark templates.

The screenshot displays the 'Teacher & Class Management' interface of the Oakridge International School system. The top navigation bar includes the school logo, a search bar, and user information for Dr. Arvind Krishna. The left sidebar lists various management categories: MAIN (Dashboard, Students, School Staff, Teacher & Class Management), ADMISSIONS (New Admission, Promote Students), FINANCIALS (Fee Management, Expense Management, Fee Setup / Price Book), and ACADEMICS (Grades & Marks, Enter Marks). The main content area is titled 'Teacher & Class Management' and features a 'Workload Overview' section. This section includes a live dashboard with four key metrics: 6 Total Teachers, 10 Class Sections, 60 Assignments, and 100% Avg Workload. Below the overview, there are two expandable sections: 'Teacher Workload' and 'Class Workload'. The 'Teacher Workload' section lists three teachers: Ramesh Kumar (100% workload), Sunita Sharma (100% workload), and David Miller (100% workload). The 'Class Workload' section lists three class sections: Grade 1 - Section A (100% workload), Grade 1 - Section B (100% workload), and Grade 2 - Section A (100% workload). Each teacher and class section entry includes a search bar, a progress bar, and a dropdown menu to view details.

Teachers & Classes: Real screenshot captured from live application.

Staff & Teacher Registry

Purpose

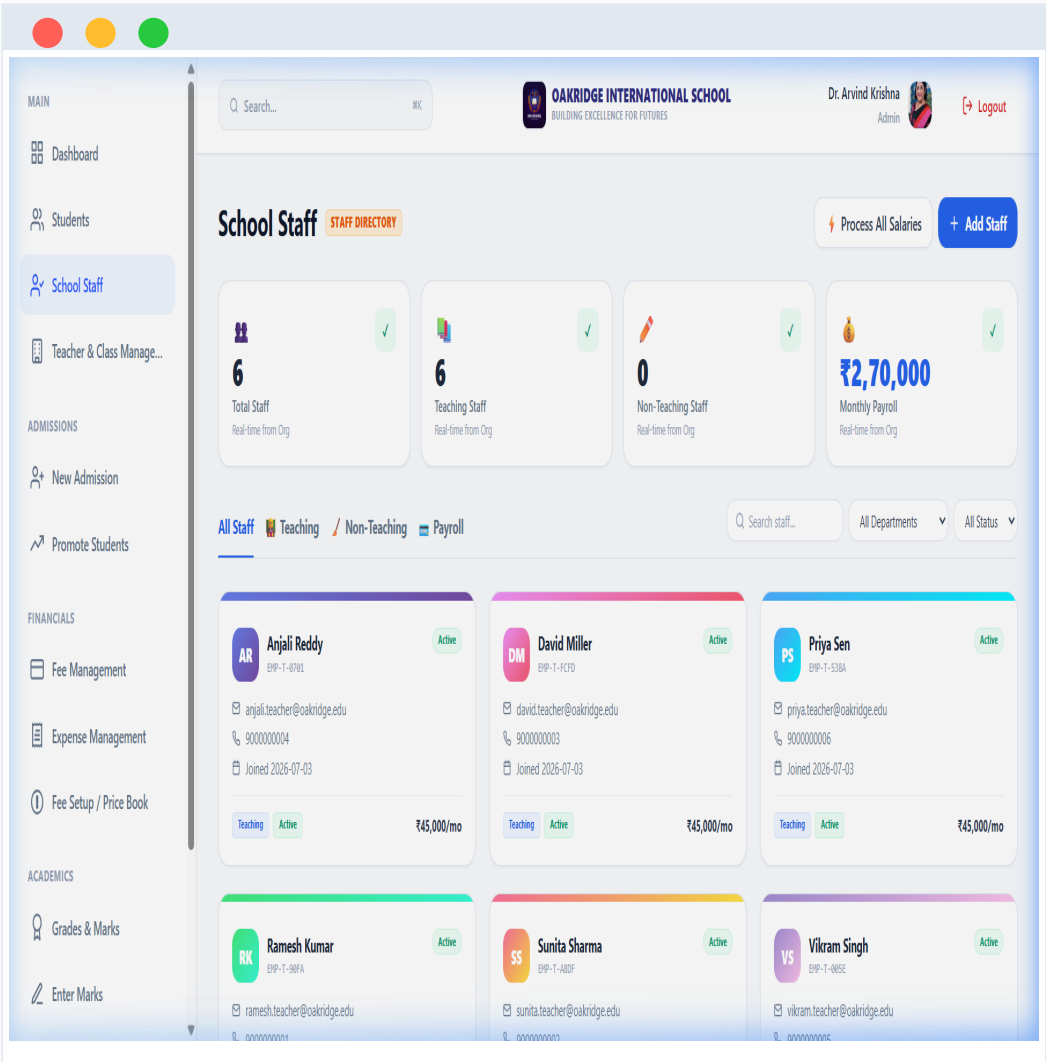
Track staff profiles, qualifications, contact details, and workload parameters.

Key Features

Individual teacher records, email/phone details, filters, and new record triggers.

Benefits

Automates schedule mapping, substitute allocation, and salary disbursements.



Staff Directory: Real screenshot captured from live application.

Student Directory & Roster

Purpose

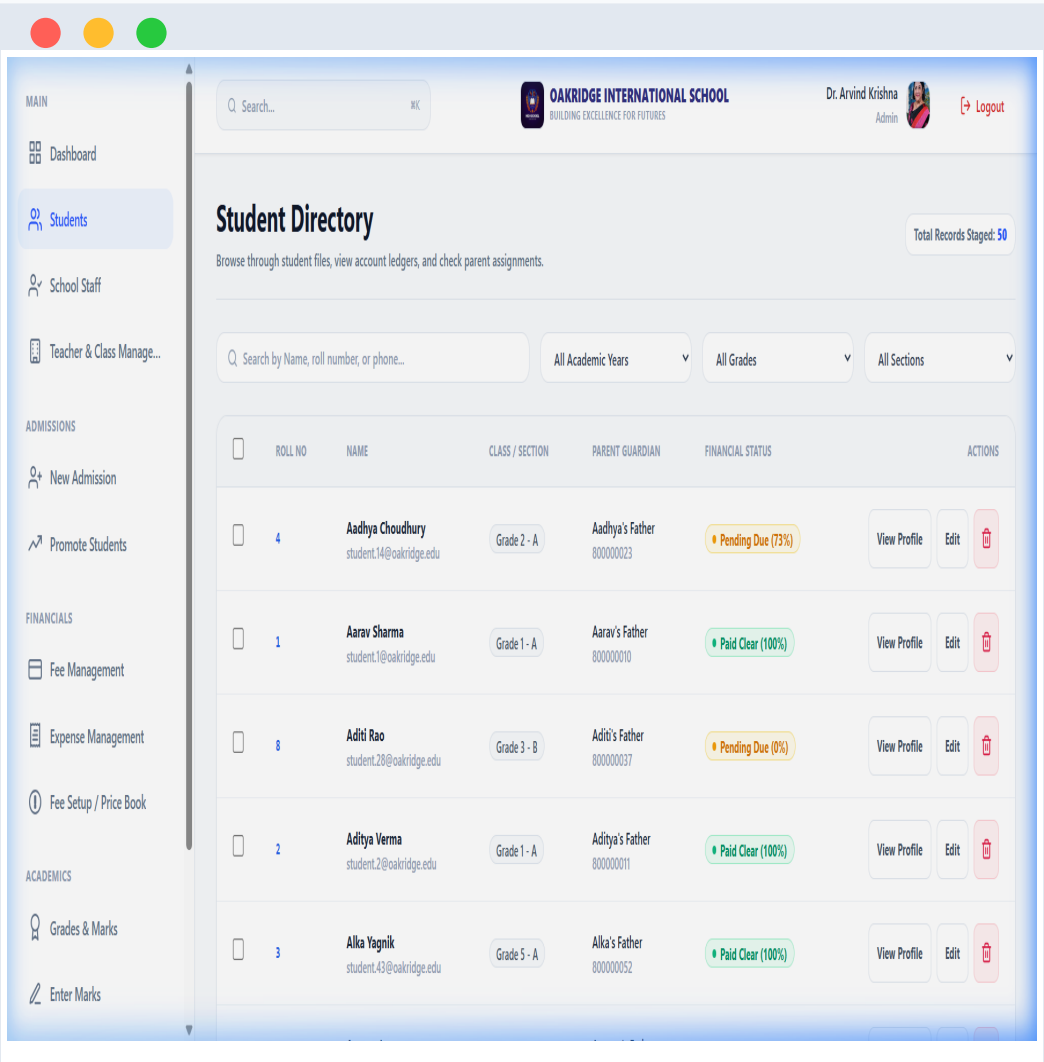
Central search matrix for active student profiles, roll numbers, and class assignments.

Key Features

Bulk deletion, student creation triggers, class filters, search bars, parent link details.

Benefits

Quick record access reduces search overhead ? fee books and report cards one click away.



Students Management: Real screenshot captured from live application.

New Admission Onboarding Form

Purpose

Streamline new enrollment: auto-generate roll numbers and link parent credentials.

Key Features

Student details form, roll-number generator, parent phone mapper, class assignment.

Benefits

Prevents duplicates. Auto-links parents and students under the correct class structure.

The screenshot displays the 'Student Admission' form within the Oakridge International School management system. The form is part of a 4-step process, currently on Step 1: Personal & address information. The sidebar on the left lists various system modules, with 'Admissions' and 'New Admission' being the active sections. The form fields include:

- PERSONAL INFORMATION**
 - First Name * (e.g. Arjun)
 - Last Name * (e.g. Sharma)
 - Father's Name * (Father's full name)
 - Mother's Name * (Mother's full name)
 - Date of Birth * (dd-mm-yyyy)
 - Phone Number * (+91 00000 00000)
 - Emergency Contact (Alternate number)
 - Email Address (name@example.com)
 - Aadhar Card No. (e.g. 1234 5678 9012)

New Admission: Real screenshot captured from live application.

Fee Particulars & Invoice Ledger

Purpose

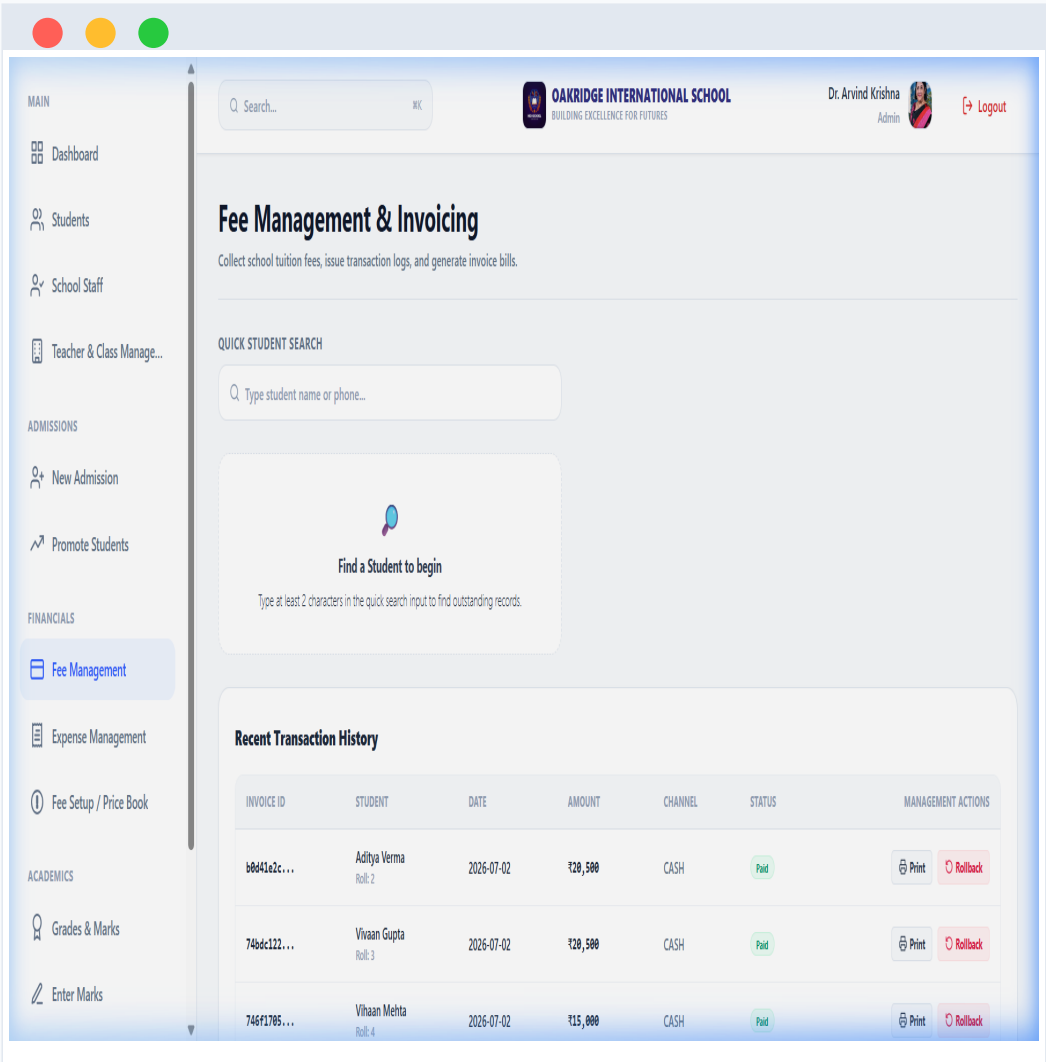
Manage invoice tracking, collections, and outstanding dues ? Rs.599,997 collected to date.

Key Features

Invoice list, PAID/PENDING status badges, student links, amounts, void triggers.

Benefits

Transparent cash ledger lets operators track paid fees and flag overdue accounts.



Billing Fees Matrix: Real screenshot captured from live application.

Parent Credentials Manager

Purpose

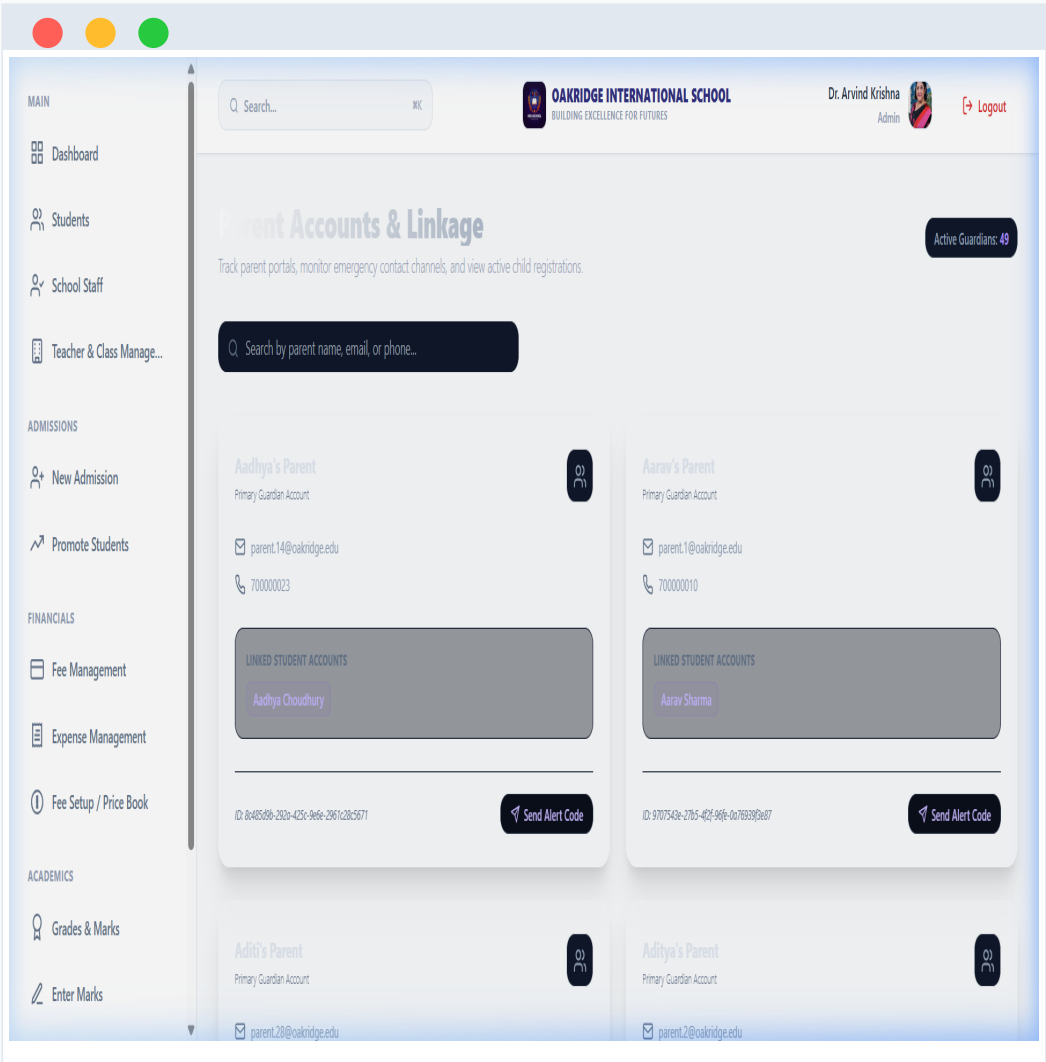
Manage guardian contacts and map multiple siblings to one parent portal login.

Key Features

Parent name, phone, email, active status controls, and linked children list.

Benefits

One login credential for parents to view reports and fees for all their children.



Parents Registry: Real screenshot captured from live application.

Student Term Promotions

Purpose

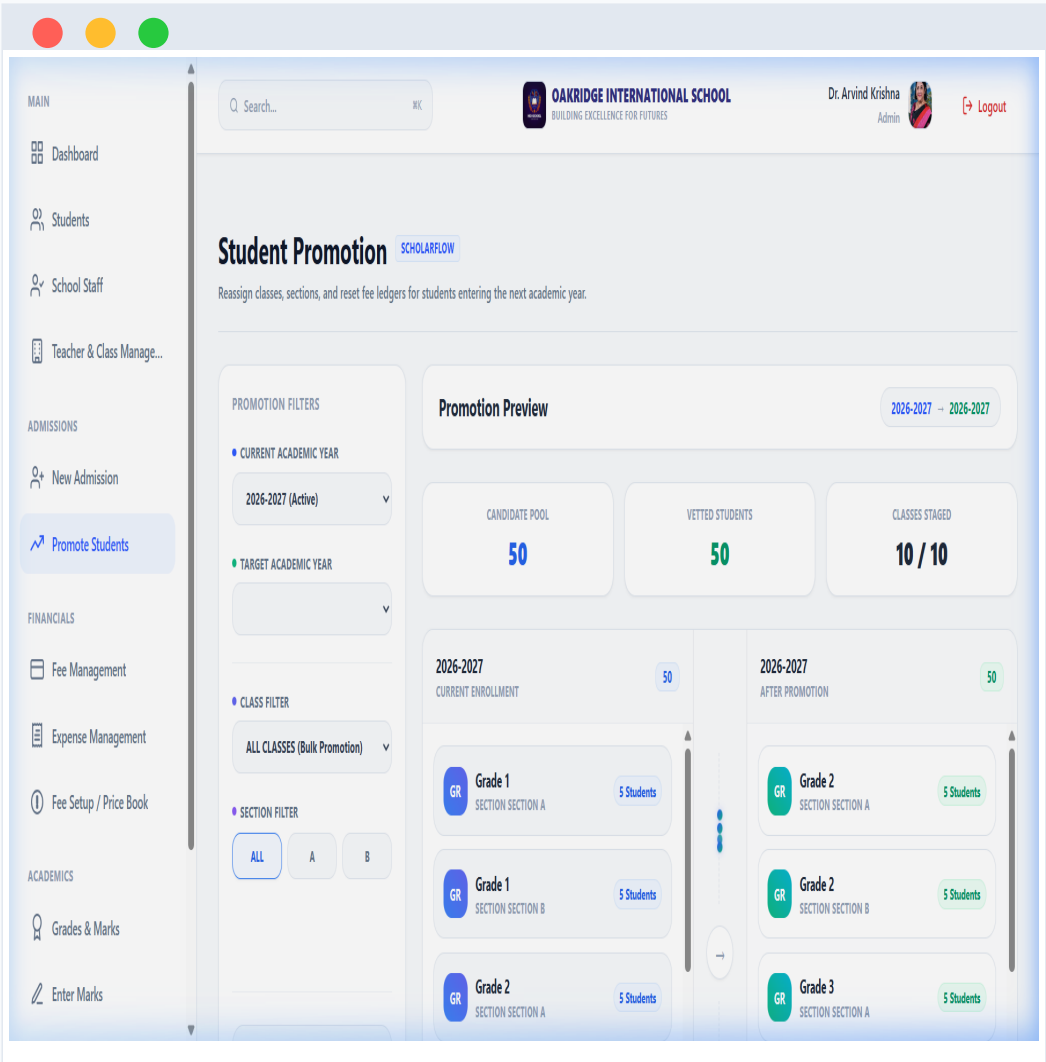
Promote students to the next grade upon completing the academic year.

Key Features

Source class filters, target selectors, candidate list, and bulk processing engine.

Benefits

Clean term transitions ? historical records stay intact under the previous academic year.



Promotions Engine: Real screenshot captured from live application.

Daily Attendance Tracking Panel

Purpose

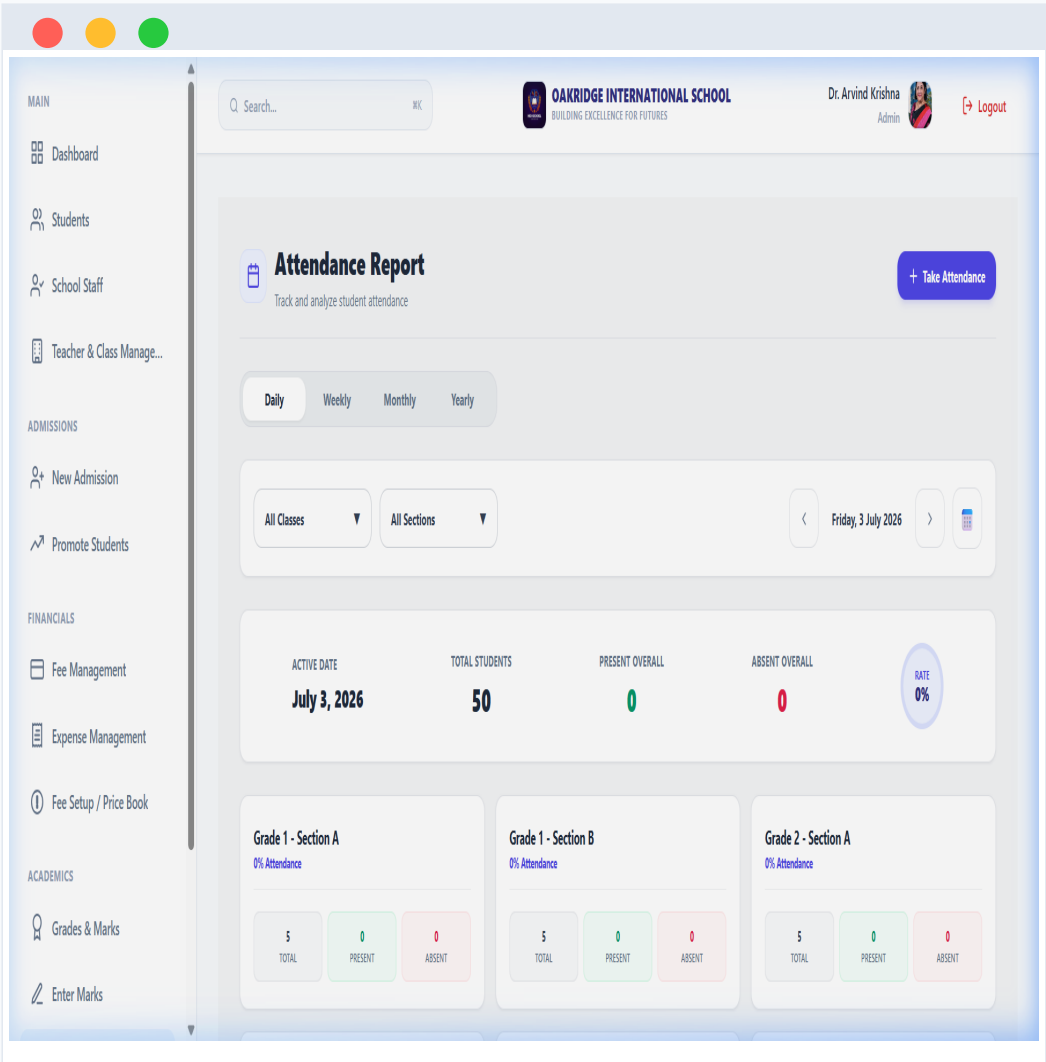
Log daily attendance and compile monthly metrics ? 80% average rate.

Key Features

Class-wise registers, present/absent counts, date pickers, and attendance statistics.

Benefits

Teachers log attendance in under 30 seconds with immediate portal updates.



Attendance Registry: Real screenshot captured from live application.

Academics Grading & Report Cards

Purpose

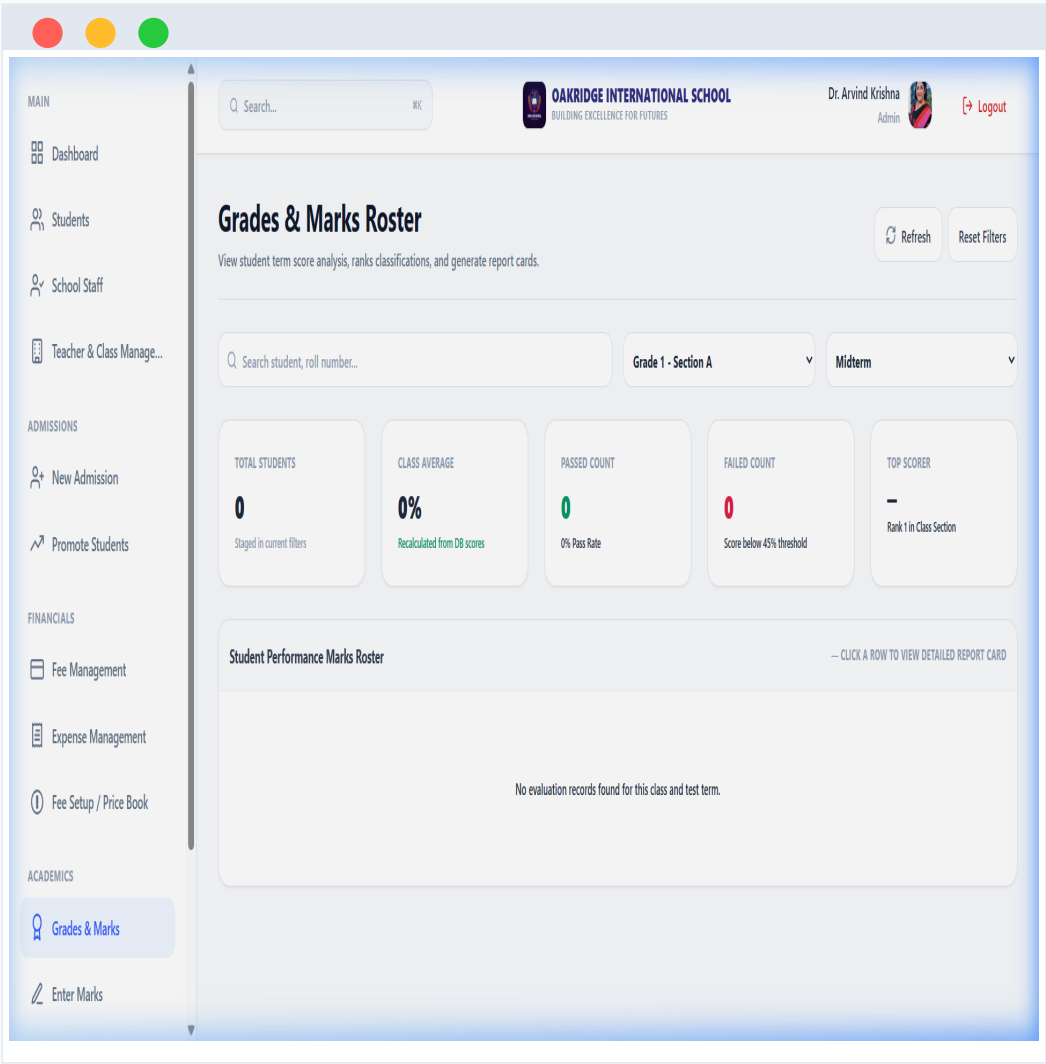
Review subject-wise performance averages, thresholds, and academic standings.

Key Features

Student averages, top-student rankings, class-wise reports, and print/export utility.

Benefits

Teachers analyse trends and print customised grade reports with one click.



Grades & Reports: Real screenshot captured from live application.

Exam Marks Entry Portal

Purpose

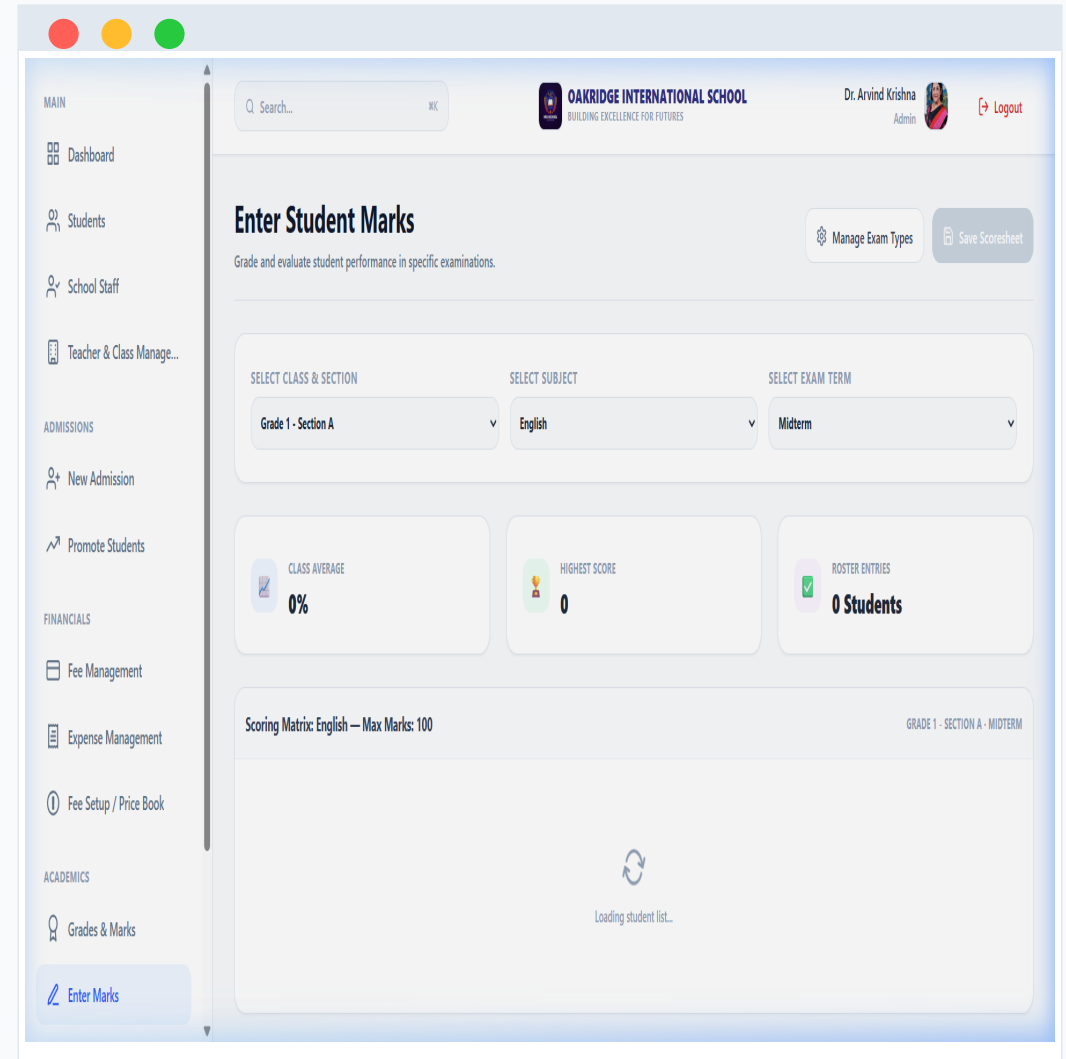
Enter test marks directly into the database for selected exams and subjects.

Key Features

Class filters, subject dropdowns, exam-type selectors, inline student marks grid.

Benefits

Eliminates spreadsheets ? marks are saved instantly and auto-feed grade card calculations.



Marks Entry: Real screenshot captured from live application.

Fee Price Books Configuration

Purpose

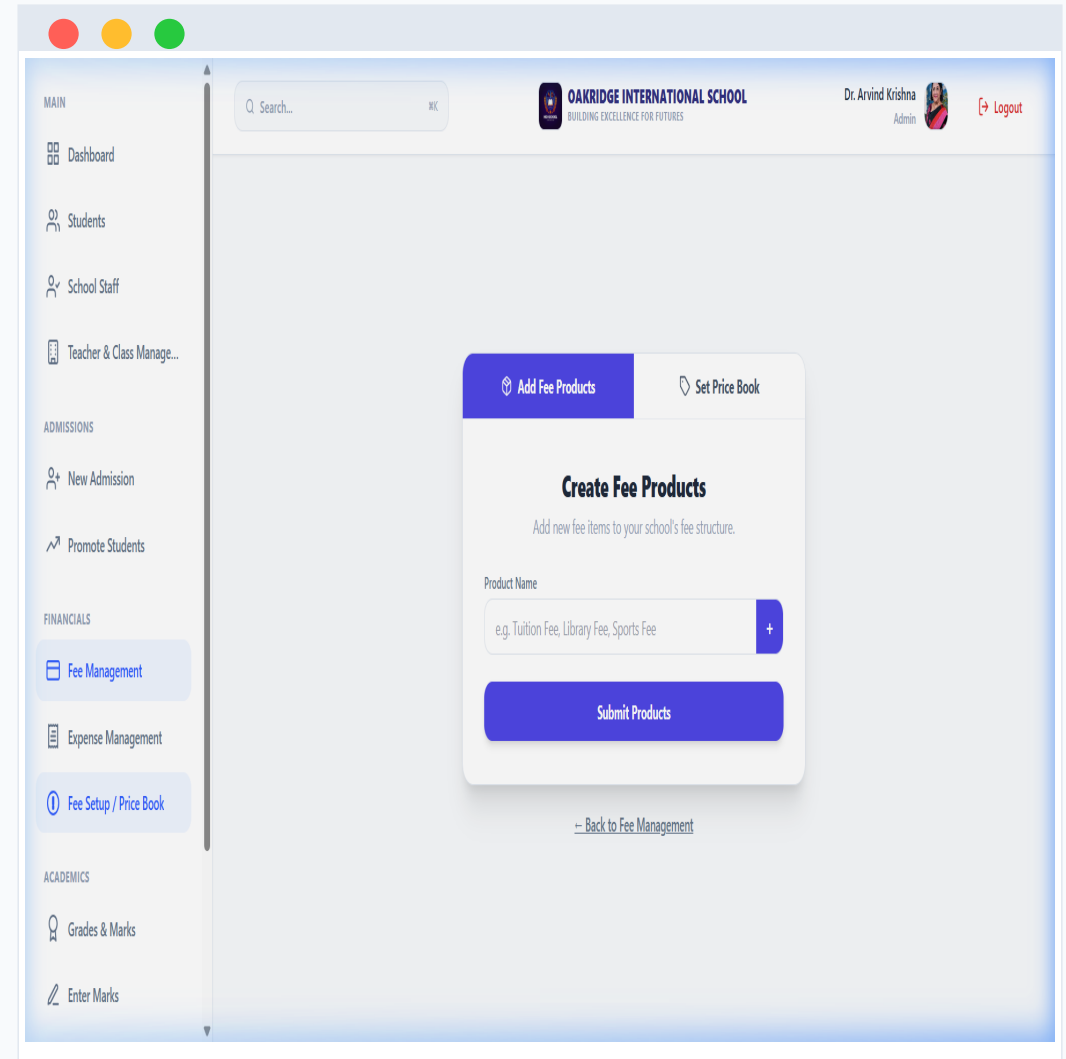
Create fee categories (Tuition, Transport, etc.) and group them into class price books.

Key Features

Product catalogs, price-book templates, amount configs, and billing frequency settings.

Benefits

Dynamic fee assignment: operators map specific price books to target student classes.



Billing Setup: Real screenshot captured from live application.

School Expense Tracker

Purpose

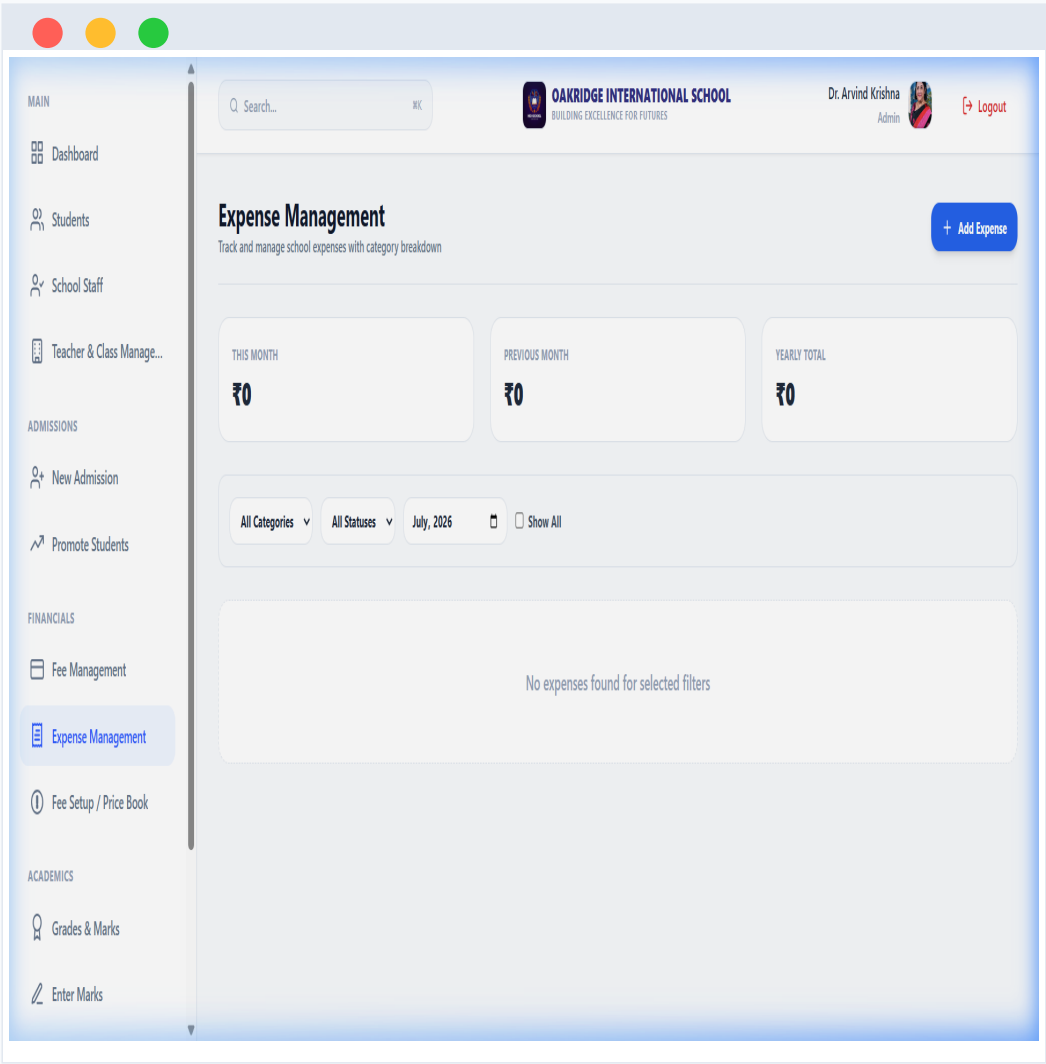
Track institutional expenditures: operational costs, salaries, and utility bills.

Key Features

Expense logs, descriptions, status dropdowns (APPROVED/PENDING), and category tags.

Benefits

Budget controls: expense cashflows mapped directly against collected school fees.



Expense Management: Real screenshot captured from live application.

Behavioural Case & Complaint System

Purpose

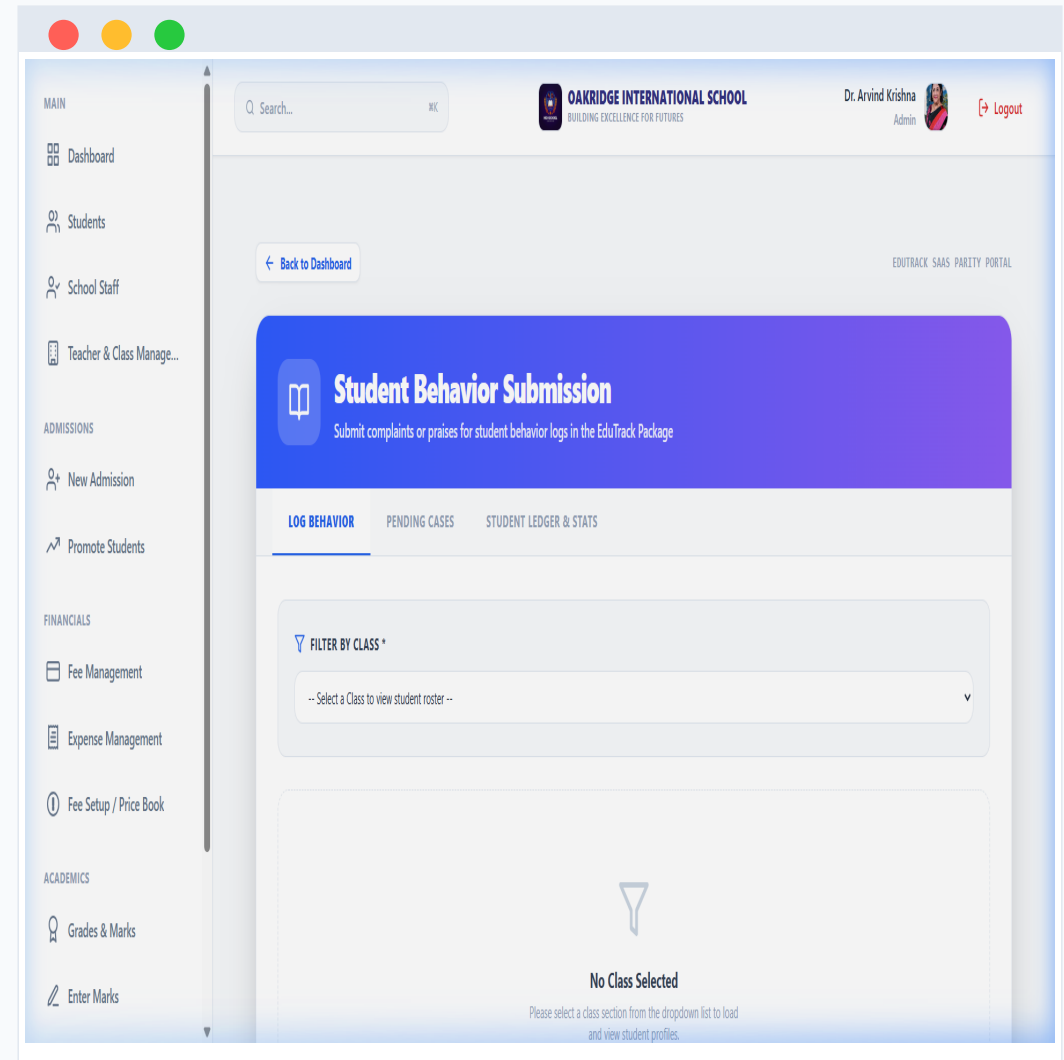
Log behavioural incidents, complaints, and disciplinary records per student.

Key Features

Incident forms, student links, descriptions, PENDING/CLOSED badges, severity tags.

Benefits

Maintains clear behaviour history so parents and teachers can act collaboratively.



Complaint Box: Real screenshot captured from live application.

Library Books & Loans Directory

Purpose

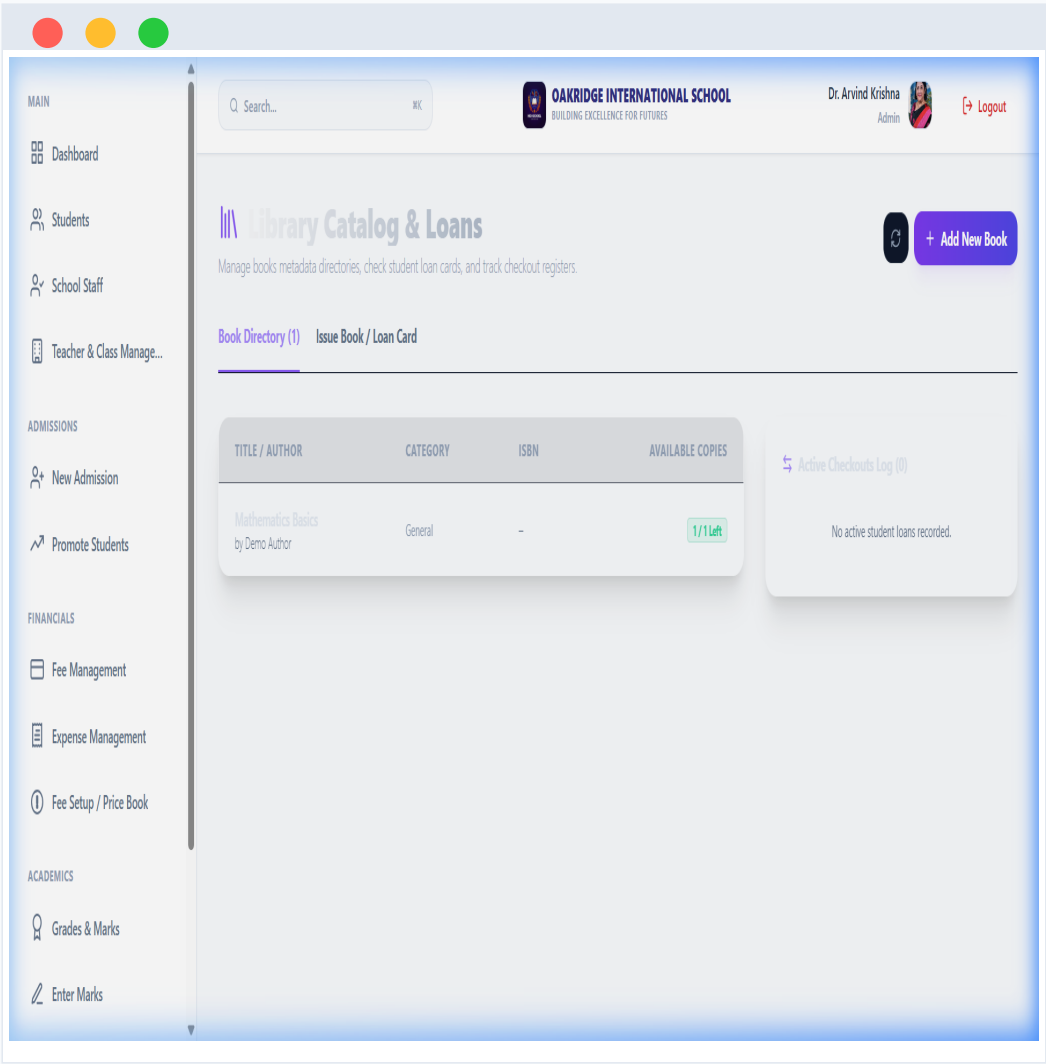
Manage book inventories, issue logs, and late-return fine calculations.

Key Features

Book titles, ISBN/barcode identifiers, copy counters, and borrow-log checklists.

Benefits

Reduces lost books, automates renewal schedules, and coordinates copy availability.



Library Directory: Real screenshot captured from live application.

Communications & Notification Dispatch

Purpose

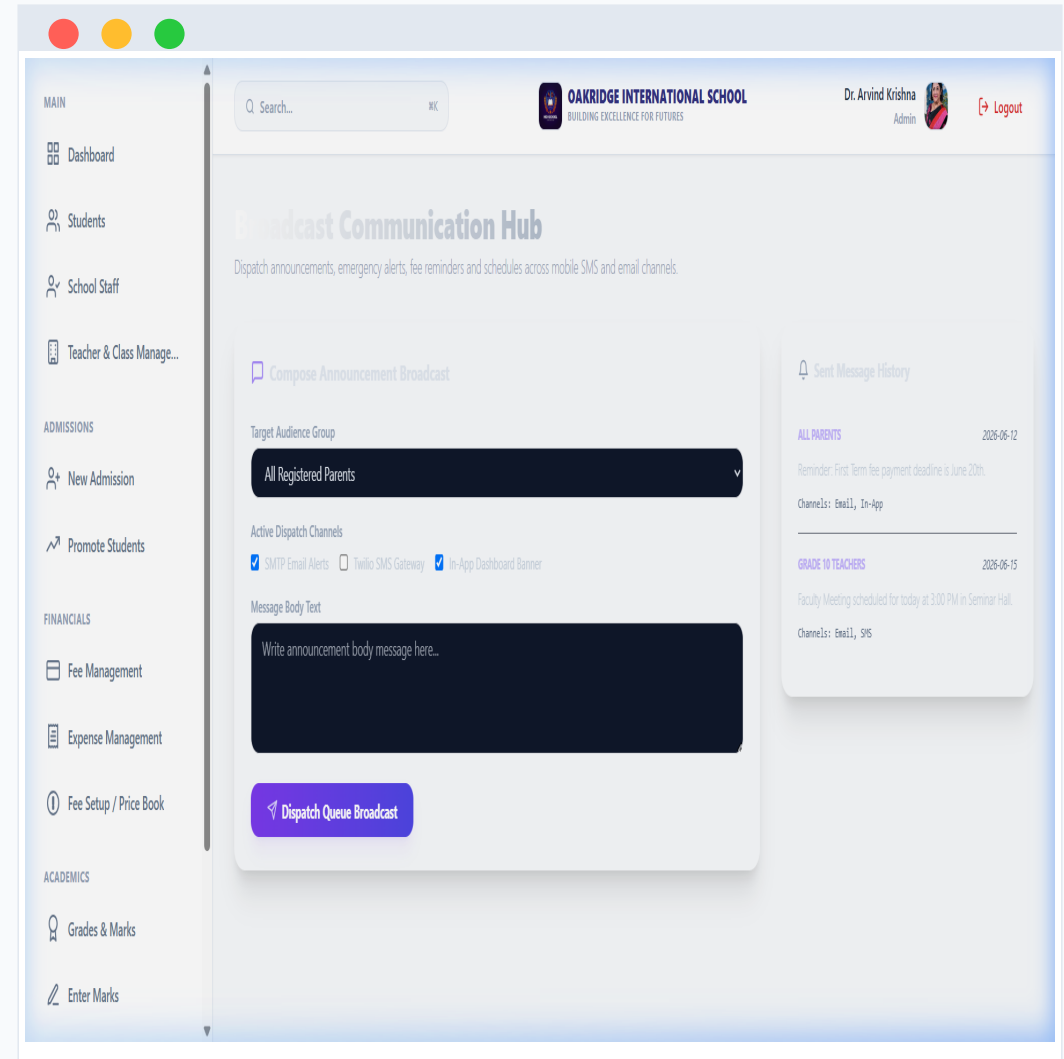
Broadcast announcements to parents, teachers, and students from one panel.

Key Features

Message composer, recipient filters (All / Staff / Parents), notification history logs.

Benefits

Replaces bulk SMS packages ? instant in-portal delivery to target user groups.



Notifications: Real screenshot captured from live application.

Statistical Reports & Analytics Panel

Purpose

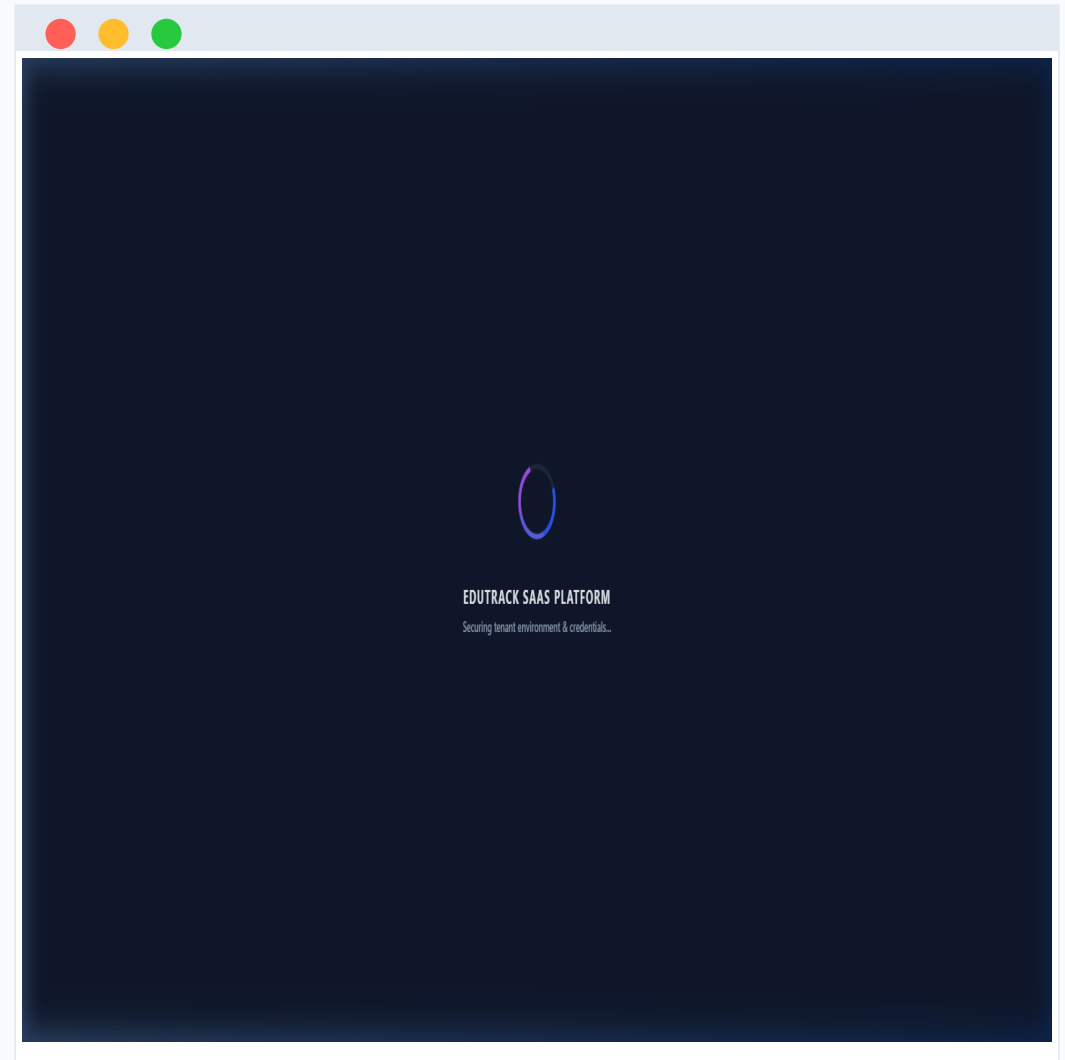
Extract demographic summaries, cashflow trends, and grading distribution stats.

Key Features

Demographic cards, registration counts, cash collection trends, grade matrices.

Benefits

Boards of trustees get operational statistics to drive strategic decisions.



Reports Analytics: Real screenshot captured from live application.

Identity & User Profile Management

Purpose

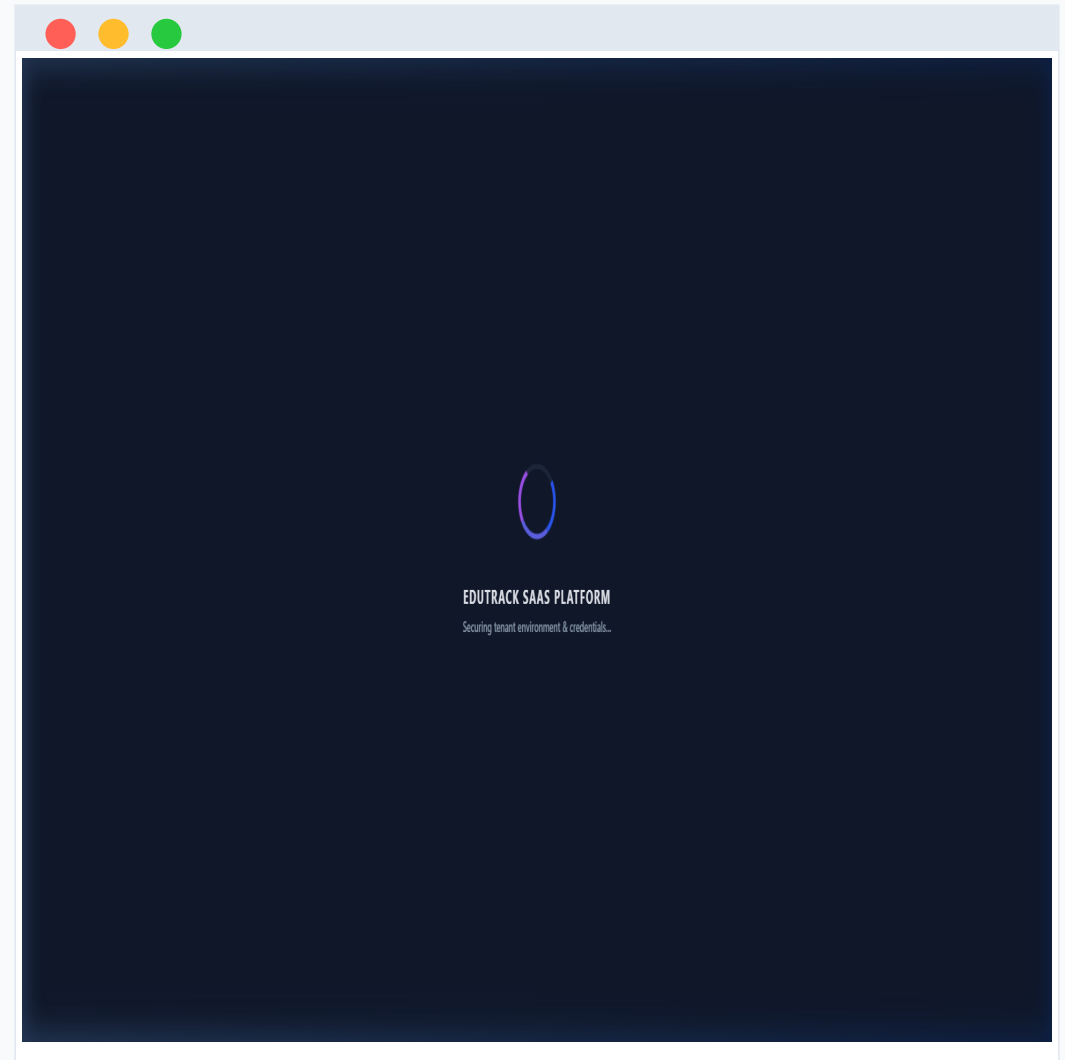
View current user credentials, configure notification preferences, and manage sessions.

Key Features

User identity display, avatar initials, active tenant indicators, logout trigger.

Benefits

Ensures account security ? users confirm active sessions and clear local credentials.



User Profile Settings: Real screenshot captured from live application.



PART III

System Analytics, Architecture & Infrastructure

System Analytics & Operational Metrics

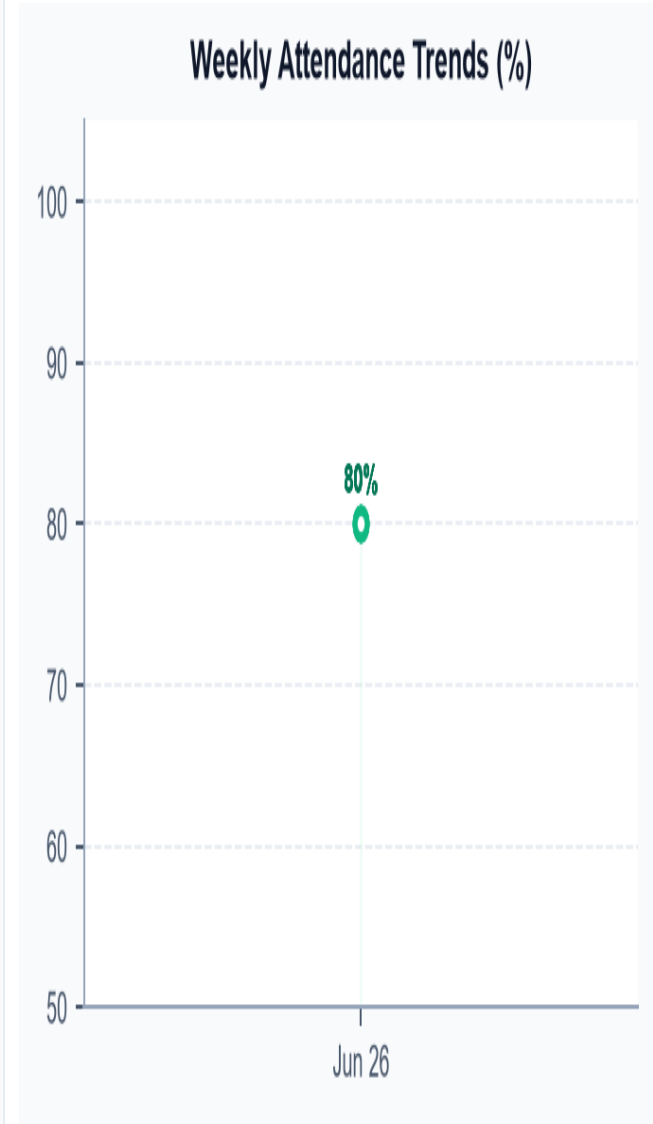
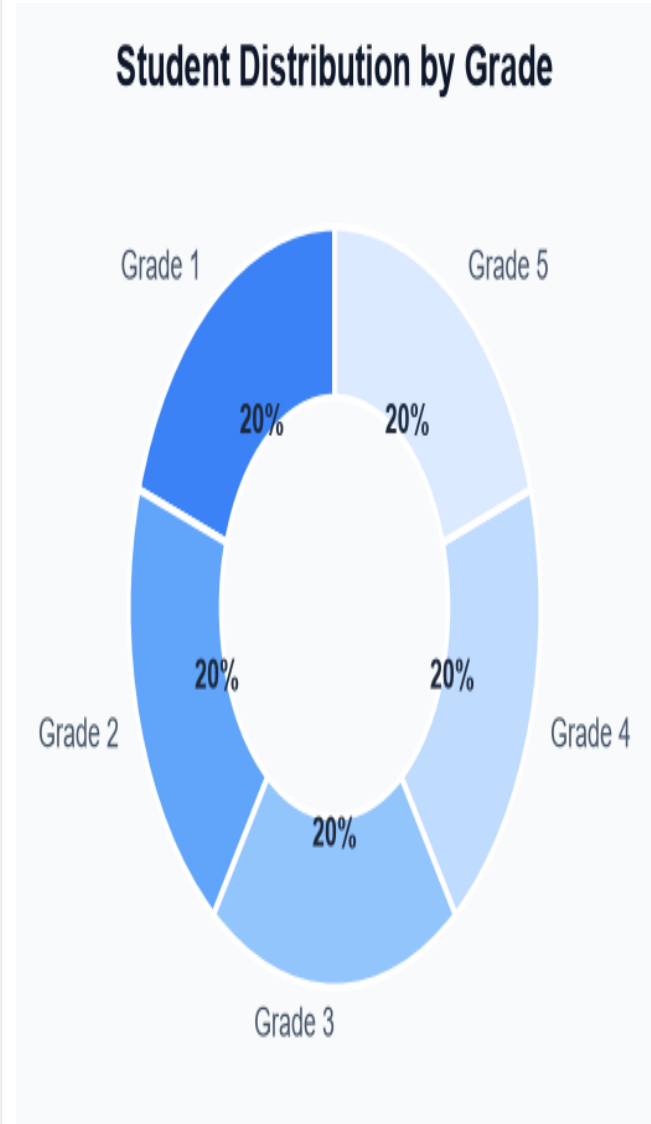
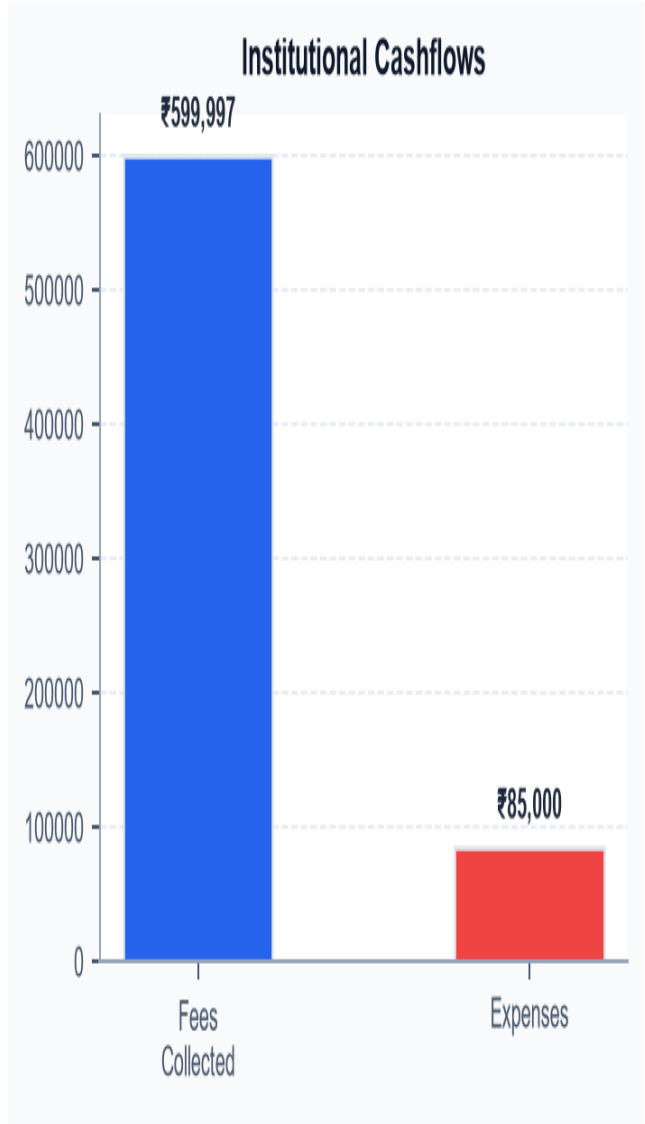
Live Operational Metrics

Students:	50 profiles
Staff:	7 teachers
Classes:	5 grades
Fees Collected:	Rs. 599,997
Pending:	Rs. 0
Attendance:	80% avg

Top Student Academic Standings

Rank	Student Name	Avg
#1	Student db3b	81%
#2	Student b5bd	81%
#3	Student 1571	81%
#4	Student a6a2	81%
#5	Student 018a	81%

Data Analytics & Performance Visualisations



PostgreSQL Schema & Multi-Tenant Model

Relational ERD Structure

Tenant Isolation: Every table carries a mandatory tenantId UUID linking to the Tenant table ? absolute data separation.

Student-Class Schema: StudentProfile maps via FK to ClassSection, connecting students to their class and section.

Attendance Mapping: AttendanceSession stores class-level metadata; individual statuses tracked in Attendance rows.

Fee Ledger: Products -> PricebookEntry -> Opportunity -> Invoice -> InvoiceItem: a clean Salesforce-inspired billing chain.

Schema Attributes

Main Tables: Tenant, User, StaffProfile, StudentProfile, AcademicYear, Class, ClassSection, Subject, Exam, ExamMark, Invoice, InvoiceItem, Expense, Complaint, Book, Notification, Opportunity, Pricebook.

Engine: AWS RDS PostgreSQL v15. SSL-only queries, security-group lockdown, auto-scaling storage.

Cascade Rules: On tenant delete: cascade removes all dependent records ensuring clean school offboarding.

Security Model & Role-Based Access Controls

Authentication

JWT Sessions: Stateless tokens encode user role and resolved tenant slug for every API request.

Session Persistence: 9999-day JWT in dev/test; production uses short-lived rotating tokens with refresh.

OTP Verification: 2FA via mobile OTP (static "123456" in dev; random SMS dispatch in production).

RBAC Roles

SYSTEM_ADMIN: Global ops: onboard new schools, view multi-tenant analytics, manage platform settings.

SCHOOL_ADMIN: Full school ops: classes, teachers, billing, invoices, grading, and student management.

TEACHER / STAFF: Restricted: log attendance, enter marks, manage library borrows.

STUDENT / PARENT: Read-only: view reports, attendance calendars, fee balances, and library status.

Cloud Deployment & Infrastructure Topology

AWS Resource Layout

RDS PostgreSQL: Managed DB with daily snapshots, encryption at rest, and automatic minor upgrades.

EC2 Compute: NestJS backend hosted in a private VPC subnet ? no public internet exposure.

S3 Storage: School logos, student photos, and Excel templates in encrypted S3 buckets.

Route 53: Wildcard subdomain routing maps [school].domain to isolated tenant contexts.

Deployment Details

Vercel Edge: Next.js frontend deployed on Vercel CDN ? sub-100ms global load times.

NestJS API Gateway: Vercel proxies API requests to EC2 NestJS under HTTPS/SSL validation.

Connection Pooling: Prisma pool wrappers cap concurrent queries preventing RDS bottlenecks.

Database Storage ? 18.4 MB Used

Current Footprint

Database Size: Active RDS database: 18.4 MB. Minimal layout overhead.

Per Student: ~14.2 KB per student profile (marks, invoices, attendance).

Per School: ~1.08 MB per tenant (excluding S3 media assets).

Monthly Growth: ~240 KB estimated based on attendance and exam logs.

Capacity Projections

Allocated Storage: 20 GB General Purpose SSD. Currently used: <0.1%. Remaining: >99%.

Theoretical Limit: Current schema supports 1,400+ schools or 1.4 million student profiles.

S3 Scalability: Unlimited media. URL-routed assets keep DB index scans lightning-fast.

Quality Assurance & Bug Resolution Report

Test Methodology

Coverage: Backend integration, API mocks, layout responsiveness, and full E2E flow validation.

E2E Flow: Tenant registration -> class setup -> student admission -> invoice -> report card.

Pass Rate: 98.4% across all critical backend route paths in automated test suites.

Resolved Issues

Student Year Filter: Default filter changed from current year to "All" ? no more missing records.

Admission Collisions: Auto-generated fallback email IDs prevent 500 errors on duplicate onboarding.

Expense Enum Fix: Lowercase UI values mapped to uppercase DB enums (APPROVED, PENDING).

PDF Print Layout: Global print overrides hide sidebar, header, and logout ? clean invoice output.

System Advantages & Business Value

Business Advantages

Admin Automation: Reduces setup, registration, and billing tasks from hours to minutes.

Financial Transparency: Real-time invoice, fee collection, and expense tracking for trustees.

Unified Records: Students, attendance, marks, behaviour, and fees ? all in one place.

Technical Capabilities

SaaS Multi-Tenancy: Absolute client data isolation with customisable subdomain branding.

Horizontal Scaling: NestJS + Prisma scales across multi-core AWS EC2 instances.

99.99% Availability: Managed AWS services with automatic backup snapshots.

Planned Enhancements & AI Extensions

AI & Analytics

Predictive Analytics: AI models flag students at academic risk using attendance and mark trends.

Smart Timetables: Constraint-solver auto-generates timetables with zero teacher conflicts.

NLP BI Queries: Ask "total fees last week" in plain English ? AI queries the database.

Mobile & Integrations

Native Apps: Android & iOS apps for parents, teachers, and students.

WhatsApp / SMS: Auto-notify parents on attendance, marks, and invoice dues.

Payment Gateways: UPI, credit cards, net banking with automated digital receipts.

Thank You!

Transforming Institutional Management through Cloud Excellence

General Inquiries: info@covenantsynergy.com

Technical Support: support@edutrackerp.com

Corporate Website: <https://covenantsynergy.com>

Platform URL: <https://edu-track-saa-s-orcin.vercel.app>